

PySR on Fully Observable NeuronBench

Six deterministic worlds · 3 seeds · max_evals=1,000,000 · generated 2026-08-18

Reduction. The original task combines active experiment design with hidden-state dynamical system discovery. Here each scalar SR problem exposes I_{ext} , membrane voltage V , and every channel open fraction ϕ_c . PySR learns the exact current-balance vector field $dV/dt = I_{\text{ext}} - \sum_c g_c \phi_c (V - E_c)$. Training and test points are independent scrambled Sobol collocation states spanning the benchmark's current/voltage range. No observation noise, numerical differentiation, active learning, or latent-state inference remains.

Search. Both methods use precisely the same data, seeds, $+/-/*$ function set, 15×33 populations, max size 35, float64 arithmetic, and evaluation cap. The evolved condition replaces PySR's mutation, selection, survival, and loss with the best training-score bundle from run 538190. The target is divided by its training RMS during search and exactly unscaled for all reported physical-unit errors.

Recovery rule. A frontier counts as recovered when at least one equation has held-out global NRMSE $\leq 1e-6$; near-exact is $\leq 1e-3$ and close is ≤ 0.05 . Exact symbolic equality is reported separately and is stricter because fitted floating constants rarely simplify identically. Affine-calibrated NRMSE is a secondary shape metric mirroring 538190's custom loss; it does not count as full dynamics recovery because incorrect scale/offset remain incorrect dynamics.

Aggregate result

method	runs	recovered	exact	median raw	median calibrated
vanilla PySR	18	1	0	4.09e-03	4.02e-03
evolved PySR (538190)	18	0	0	4.63e-03	3.51e-03

Across 18 paired runs, evolved wins 9, baseline wins 9, ties 0; median evolved/baseline NRMSE ratio = 0.967 (< 1 favors evolved).

Verdict: 538190 does not clearly improve full equation recovery. Vanilla is the only method with a strict recovery (1 vs 0) and has the lower raw median NRMSE (4.09e-03 vs 4.63e-03). The paired result is exactly split 9-9. The evolved method is modestly better only on the secondary affine-calibrated shape metric (3.51e-03 vs 4.02e-03), consistent with the affine-invariant loss it evolved.

Sources: <https://arxiv.org/abs/2608.09696>
[@c354622458c460b419cab821d482c879f0578377](https://github.com/murphyk/neuronbench)
Evolved algorithm: runs/538190

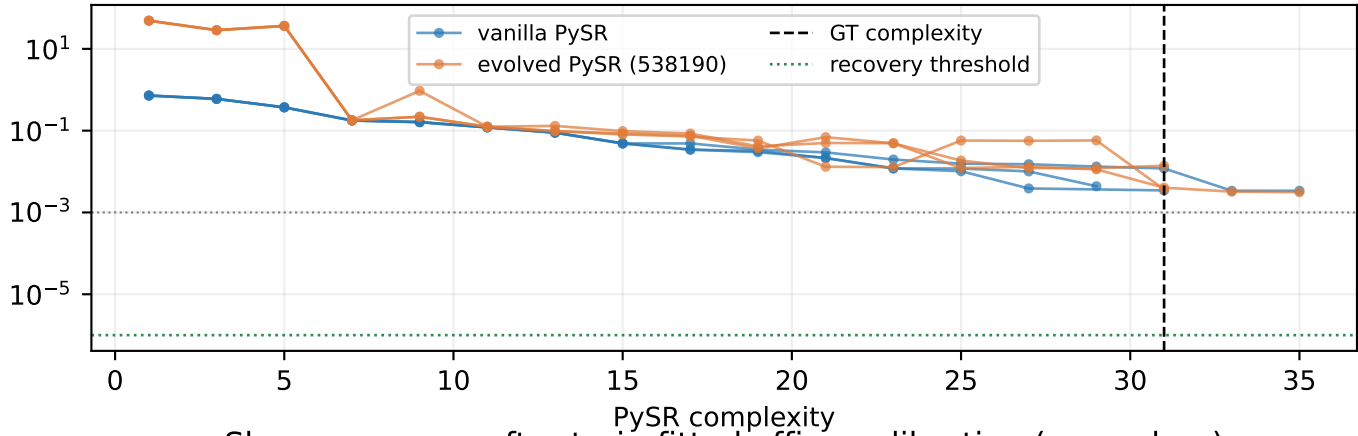
z rebound

low-threshold inward Z, de-inactivated by hyperpolarisation -> depolarisation block

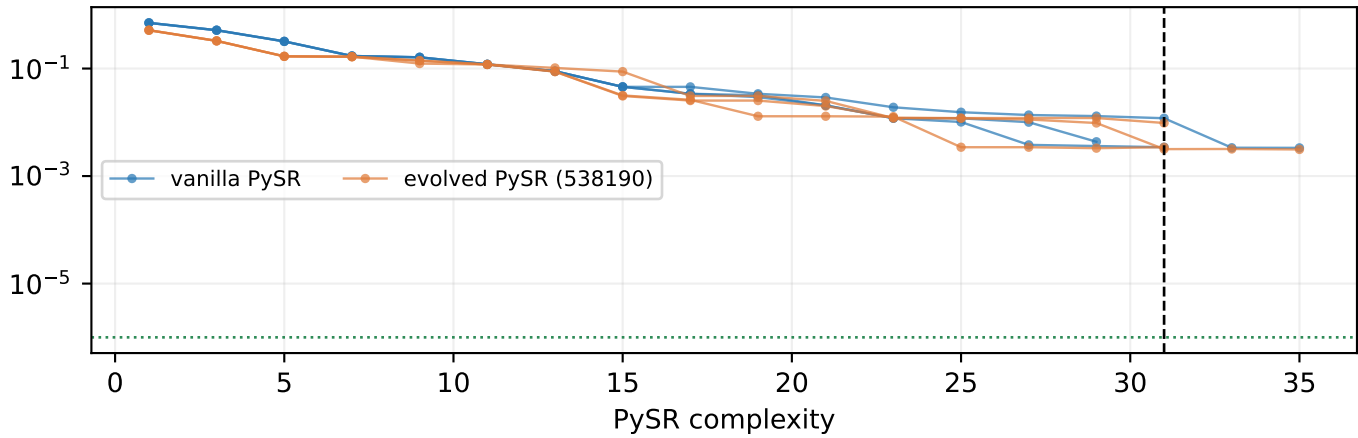
Ground truth (reference complexity 31):

```
i_ext - (120)*phi_na*(v - (50)) - (36)*phi_k*(v - (-77)) - (0.29999999999999999)*(v - (-54.399999999999999)) - (4)*phi_z*(v - (120))
```

Discovered Pareto frontiers (all seeds)



Shape recovery after train-fitted affine calibration (secondary)



vanilla PySR

seed 0: close; NRMSE=4.379e-03, R²=0.99997147, complexity=29, best calibrated=4.347e-03, exact=False, runtime=110s
(5644.415210490356)*(phi_k*(-0.4955009162062544) + phi_na*(3.2000333911271386 - v)*0.0212335671269716 + phi_na + phi_z*0.07669546712411542 - 0.0474793667519685*(phi_k*0.13604862309733262 + phi_z*0.015351285968024582))

seed 1: close; NRMSE=3.385e-03, R²=0.99998295, complexity=35, best calibrated=3.347e-03, exact=False, runtime=106s
(5644.415210490356)*(phi_na - (-0.10017751766223448)*phi_z + (phi_k - (-0.06971734982079879)*phi_na*v + (-0.011896449415342604*v - 0.2828598366344362)*(phi_k + phi_k + (phi_na + phi_z*(-0.28697893281293996))*(-0.7875321368475101))*(-0.7894209338913587))*(-0.3417170530621434))

seed 2: close; NRMSE=3.457e-03, R²=0.99998222, complexity=31, best calibrated=3.443e-03, exact=False, runtime=66s
(5644.415210490356)*(v*(-0.021276695377504172)*phi_na + phi_na*1.0607456152297112 + (phi_k - ((phi_k + phi_z*0.1227395403945439)*v*(-0.012719049535211064) + (-(-0.10077672283150999)*phi_k + phi_z)*0.16287488656532675))*(-0.5026967129916224))

evolved PySR (538190)

seed 0: close; NRMSE=3.662e-03, R²=0.99998005, complexity=31, best calibrated=3.302e-03, exact=False, runtime=86s
(5644.415210490356)*(1.0616646633803553*(phi_na - 0.020061077903904035*(-i_ext + i_ext + phi_na*v + phi_z*0.036026134360157974*v + (phi_k*(v*0.08018929282751153 + 6.160350292132949) - phi_z)*3.7468857918443588)))

seed 1: close; NRMSE=1.212e-02, R²=0.99978146, complexity=29, best calibrated=9.712e-03, exact=False, runtime=83s
(5644.415210490356)*((phi_na + (phi_k + (-(-0.3115732362652546)*phi_k*v - phi_na*(phi_z - v) + (phi_na + phi_z + phi_z)*(-2.0540269909614253))*0.04341958840348582))*(-0.4972640956886286))*1.0003288324988058)

seed 2: close; NRMSE=3.125e-03, R²=0.99998547, complexity=35, best calibrated=3.126e-03, exact=False, runtime=84s
(5644.415210490356)*((phi_na + 0.23134110858888032*(-phi_k - phi_k - 0.08660610371460493*phi_na*v - 0.3448287573497958*(-phi_z + v*(0.07559758609203311*phi_k - 0.008496641018494039*(-phi_z - 0.01923065462784559)))) - 0.15832883923649327 + 0.1420820595096272))*1.062584076667184)

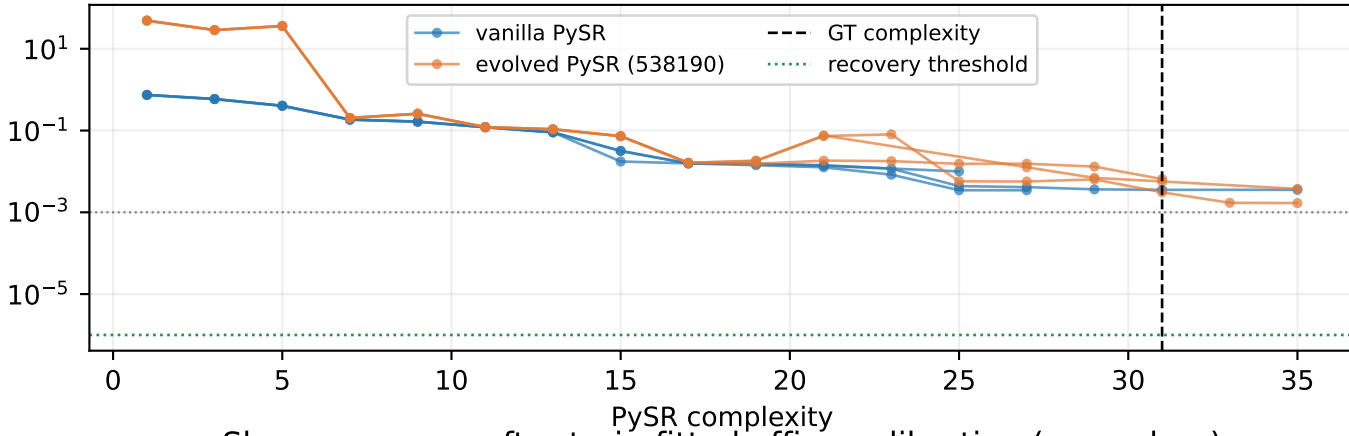
h_sag

hyperpolarisation-activated inward Ih -> voltage sag and post-inhibitory rebound

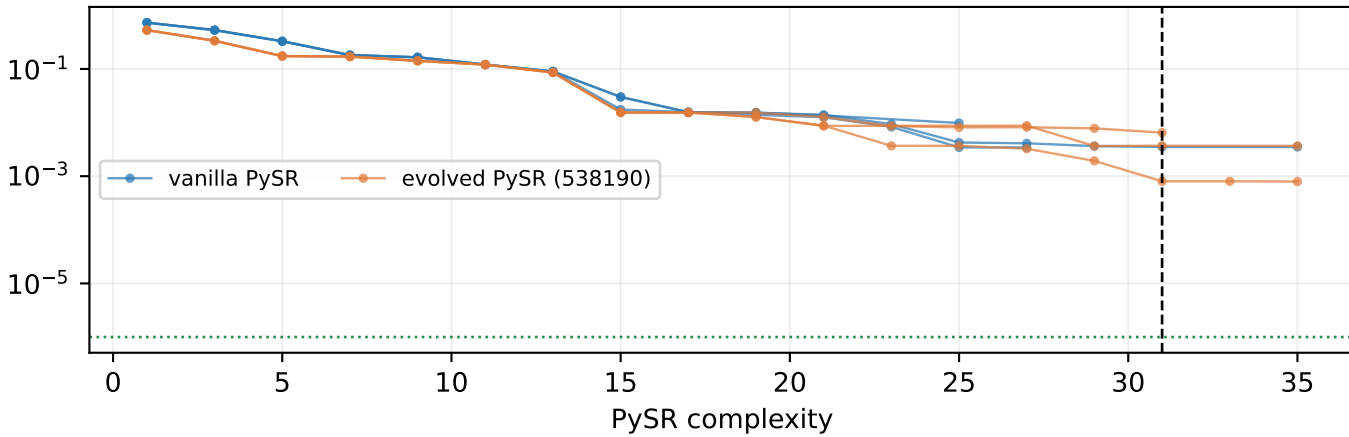
Ground truth (reference complexity 31):

$i_{ext} - (120) * \phi_{na}(v - (50)) - (36) * \phi_k(v - (-77)) - (0.29999999999999999) * (v - (-54.399999999999999)) - (5) * \phi_{ih}(v - (-30))$

Discovered Pareto frontiers (all seeds)



Shape recovery after train-fitted affine calibration (secondary)



vanilla PySR

seed 0: close; NRMSE=3.469e-03, R²=0.99998318, complexity=27, best calibrated=3.438e-03, exact=False, runtime=37s
(5488.8601710625717)*(-0.030188793163216678*phi_ih - 0.010947733671418266*v*(phi_ih*0.08612389708581782 + phi_k*0.6021332753246811 + phi_na + phi_na) + (-0.46586701838020295*phi_k + phi_na)*1.0902416665038233)

seed 1: close; NRMSE=3.550e-03, R²=0.99998238, complexity=31, best calibrated=3.512e-03, exact=False, runtime=113s
(5488.8601710625717)*(phi_na*1.0682611464675345 + (-phi_k + (phi_ih + phi_na*v - phi_na - (-0.420522163870193*phi_ih + (phi_ih*0.13836953510510988 + phi_k)*(-0.30164525520608)*v))*(-0.04323198045232606))*0.5067234799400092)

seed 2: close; NRMSE=1.001e-02, R²=0.99985998, complexity=25, best calibrated=9.822e-03, exact=False, runtime=63s
(5488.8601710625717)*(phi_k*(-0.5376928057432434) + phi_na + (-0.021976247227450868*(-phi_na + v) - 1*(-0.05563356303428238))*(phi_ih*0.029276724478445582 + phi_k*0.30307423687943086 + phi_na + 0.002838445014528097))

evolved PySR (538190)

seed 0: close; NRMSE=6.499e-03, R²=0.99994096, complexity=31, best calibrated=6.493e-03, exact=False, runtime=85s
(5488.8601710625717)*(-0.5053106801365528*phi_k + phi_na*(v*(-0.021857090111318752) - 1*(-0.09291876912254612)) + phi_na + (v*(phi_ih*(-0.10371835378886174) - phi_k - 0.02414616160188829) - (phi_ih - 1*0.0064247627830744565))*0.006573209842460466 - 0.014744781982078126)

seed 1: close; NRMSE=1.681e-03, R²=0.99999605, complexity=35, best calibrated=7.949e-04, exact=False, runtime=84s
(5488.8601710625717)*((-0.04629570910745676*phi_k + phi_na + (phi_ih + v*(phi_ih*0.0412006572720451 + phi_na) - phi_na))*(-0.020031085271642154) + (phi_ih + phi_k*v + (i_ext + phi_ih)*(-0.027237115298615715) + 0.4702807977203464)*(-0.006030290648114147))*1.0946766575682623)

seed 2: close; NRMSE=3.737e-03, R²=0.99998048, complexity=35, best calibrated=3.661e-03, exact=False, runtime=85s
(5488.8601710625717)*((-0.021903113126614335*(phi_ih + v*(phi_ih*0.0412006572720451 + phi_na) - 1*1.397351322895418) + (phi_na + phi_na + phi_na - (phi_k*((-phi_k + v)*0.018164839872090333 + 0.40860647601254735) + phi_k))*0.3637798947678655 - 1*0.034973824073868155)

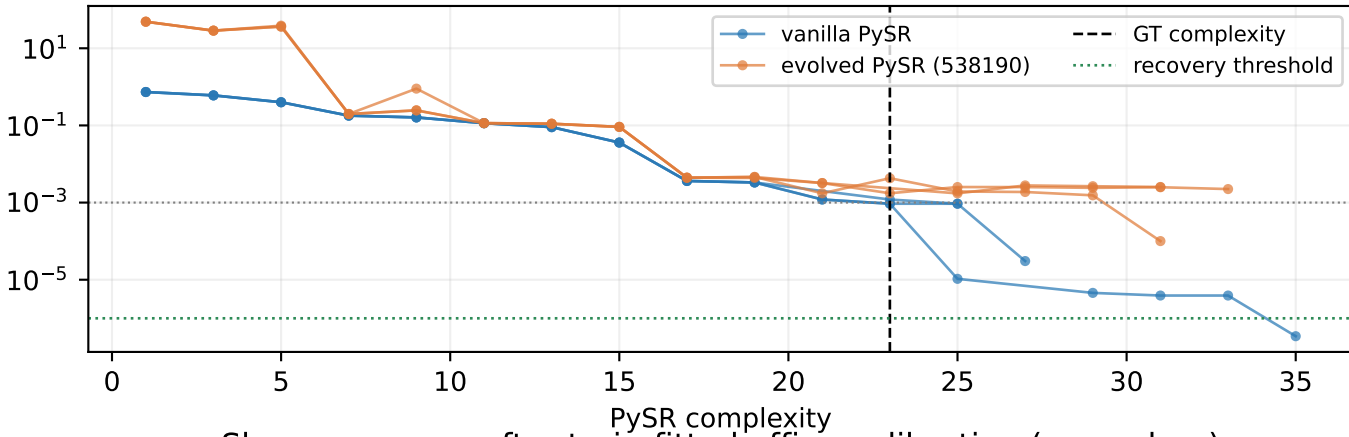
na fatigue

slow ϕ_a inactivation -> use-dependent spike-count run-down, revealed by paired long pulses

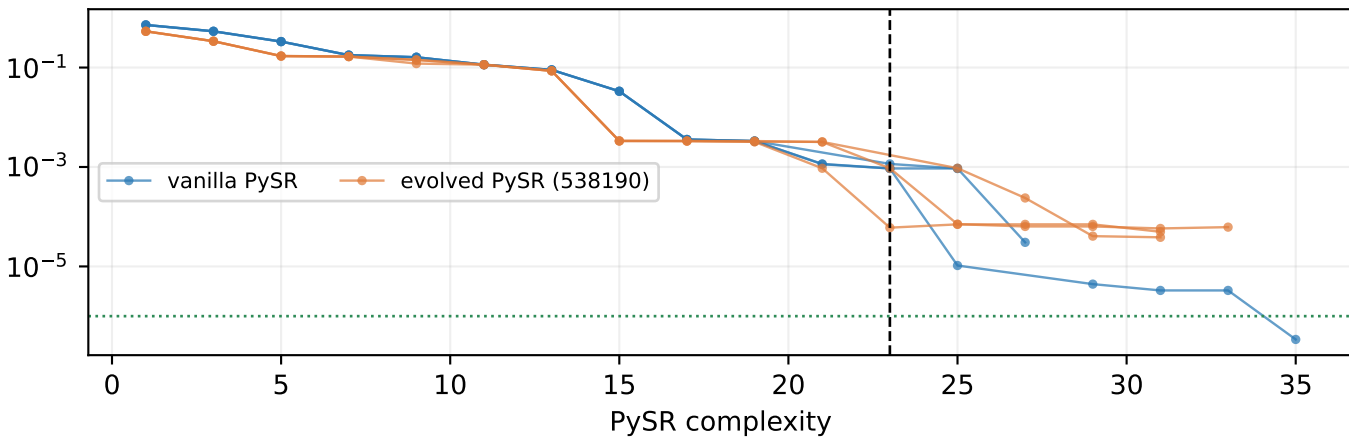
Ground truth (reference complexity 23):

$$i_{\text{ext}} - (120) \cdot \phi_a \cdot (v - (50)) - (36) \cdot \phi_k \cdot (v - (-77)) - (0.29999999999999999) \cdot (v - (-54.399999999999999))$$

Discovered Pareto frontiers (all seeds)



Shape recovery after train-fitted affine calibration (secondary)



vanilla PySR

seed 0: near-exact; NRMSE=9.342e-04, R²=0.99999876, complexity=25, best calibrated=9.342e-04, exact=False, runtime=64s

$$(5430.8722173975193) \cdot ((\phi_k \cdot (-0.46309916507761406) + \phi_a \cdot (-0.020057874560816605) \cdot v + \phi_a + (-(-0.027581984681935567) \cdot i_{\text{ext}} - \phi_k \cdot v - 0.3051622328314176) \cdot 0.0060474529456700545) \cdot 1.103966094023508)$$

seed 1: recovered; NRMSE=3.436e-07, R²=1.00000000, complexity=35, best calibrated=3.401e-07, exact=False, runtime=67s

$$(5430.8722173975193) \cdot ((\phi_k \cdot (-0.510414973113383) + \phi_a \cdot 1.104795618971783 - (v \cdot (\phi_k \cdot 0.29999929711838375 + \phi_a + 0.010834399682455037) \cdot 0.022095889021558433 + (i_{\text{ext}} - (-v + (-\phi_a - 1.405867272526661) \cdot (-9.009334219125474e-5) - 0.38331179851939)) \cdot (-0.00018414083150466697) + 0.003076031940578911))$$

seed 2: near-exact; NRMSE=3.049e-05, R²=1.00000000, complexity=27, best calibrated=3.049e-05, exact=False, runtime=62s

$$(5430.8722173975193) \cdot ((\phi_k \cdot (-0.46196556184684645) + \phi_a \cdot (\phi_k \cdot (-0.29995210831403335) - \phi_a) \cdot 0.019998228323274827 \cdot v - (-0.0001662032133774347) \cdot (i_{\text{ext}} + v \cdot (-0.31022832728575417)) - 0.00275165290995317) \cdot 1.1048195697866419)$$

evolved PySR (538190)

seed 0: near-exact; NRMSE=1.004e-04, R²=0.99999999, complexity=31, best calibrated=4.977e-05, exact=False, runtime=76s

$$(5430.8722173975193) \cdot ((\phi_k \cdot (-0.5081068595692861) - (-1.1069583748354428 \cdot \phi_a + \phi_a \cdot (-0.005503599860304827) + 0.02209405415022666 \cdot (-0.00835238318827466 \cdot i_{\text{ext}} + (v + 0.3422612607478572) \cdot (-(-0.29991885666025125) \cdot \phi_k + \phi_a - 1 \cdot (-0.002559913880018083)))) + 0.0031238533463854314))$$

seed 1: close; NRMSE=1.731e-03, R²=0.99999574, complexity=25, best calibrated=3.852e-05, exact=False, runtime=75s

$$(5430.8722173975193) \cdot ((-0.4635255038967523 \cdot \phi_k + \phi_a \cdot (\phi_k \cdot (-0.00604841888953683 \cdot (-\phi_k + \phi_a) + (-0.006416814470203829 \cdot i_{\text{ext}} + \phi_a \cdot v) \cdot (-0.026114415080603803)) \cdot 1.1023825244941605)$$

seed 2: close; NRMSE=1.741e-03, R²=0.99999569, complexity=23, best calibrated=5.807e-05, exact=False, runtime=79s

$$(5430.8722173975193) \cdot ((-0.00018473824786510221) \cdot i_{\text{ext}} + (v \cdot \phi_a \cdot (-0.009577536794878596) + \phi_a + (\phi_k + \phi_a) \cdot (v \cdot (-0.004133050038933639) - 0.31670809207686035)) \cdot 1.6133459586090007)$$

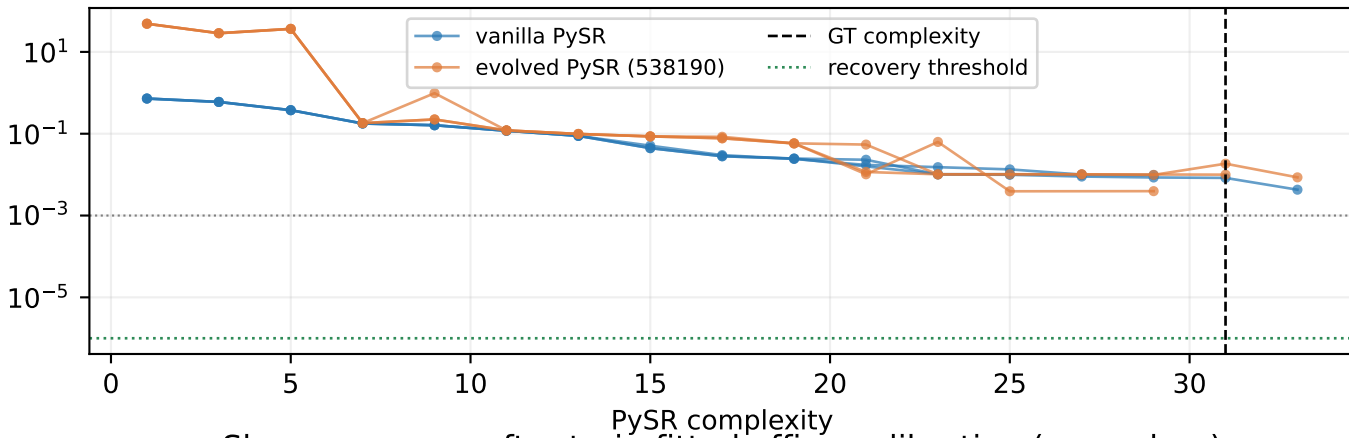
ca_rebound

fast low-threshold Ca (T-type), de-inactivated by hyperpolarisation -> rebound BURST on release

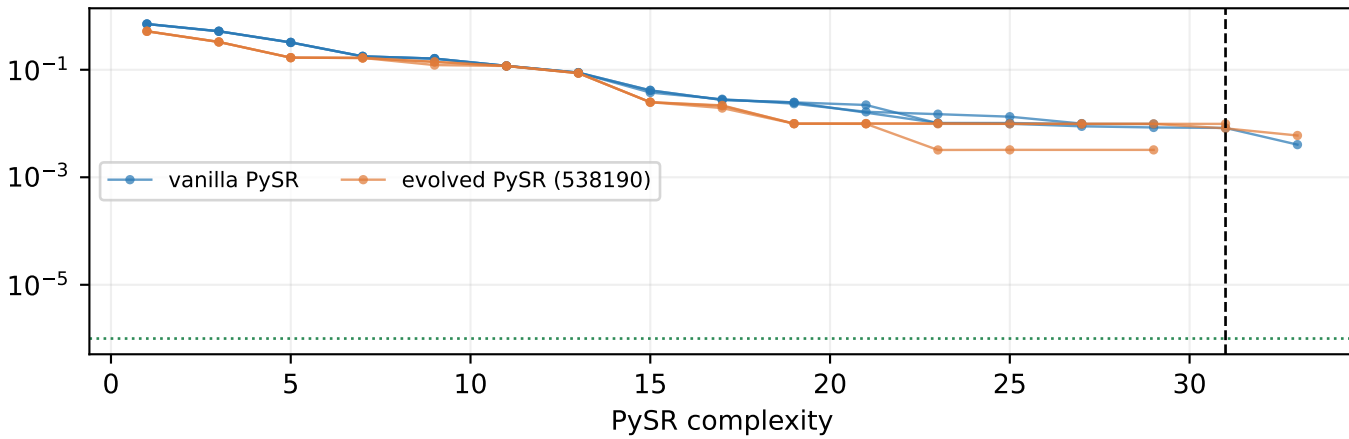
Ground truth (reference complexity 31):

```
i_ext - (120)*phi_na*(v - (50)) - (36)*phi_k*(v - (-77)) - (0.29999999999999999)*(v - (-54.399999999999999)) - (3.2000000000000002)*phi_t*(v - (120))
```

Discovered Pareto frontiers (all seeds)



Shape recovery after train-fitted affine calibration (secondary)



vanilla PySR

seed 0: close; NRMSE=1.002e-02, R²=0.99985180, complexity=27, best calibrated=9.963e-03, exact=False, runtime=62s
(5600.6894854168131)*((phi_na*(1.0624256224662099 + (phi_t - v*(phi_k*0.3091882215820917 + phi_na) + 2.4651033058549854*(-phi_k + phi_t)) - 10.33351177781065*(phi_k + phi_na))*0.02171836925443227)

seed 1: close; NRMSE=4.282e-03, R²=0.99997295, complexity=33, best calibrated=4.047e-03, exact=False, runtime=66s
(5600.6894854168131)*((phi_na*(-0.010451891595790963)*v - (-0.27481225419909744)*phi_na - (phi_t*(-0.03567252702584394) - (-0.0031522278602741944)*v*(phi_k - phi_t*(-0.044921572720261274 - 1*0.03841806356232806)) + 0.24399990712126823*(phi_k - phi_na)))*2.053439103267134)

seed 2: close; NRMSE=9.791e-03, R²=0.99985862, complexity=29, best calibrated=9.806e-03, exact=False, runtime=59s
(5600.6894854168131)*((phi_na - (-(-0.49804772967546085)*phi_k + (phi_na*v - (-0.3093237429564468*phi_k*v - (-0.006696220849703005*i_ext + (phi_na + phi_t)*(-3.197295944124287)))))*0.021717464633045806) - 0.000448326377862954)

evolved PySR (538190)

seed 0: close; NRMSE=8.577e-03, R²=0.99989150, complexity=33, best calibrated=5.986e-03, exact=False, runtime=80s
(5600.6894854168131)*(phi_t*(phi_t + v*(-0.01058590308831275))*0.06121875932658907 + (-phi_k + phi_na*(0.19230539675229152 + v*(-0.043524997630348686)) + phi_na + phi_na + (phi_k*v + phi_na)*(-0.013027543986683552))*0.4955487935979659)

seed 1: close; NRMSE=3.917e-03, R²=0.99997737, complexity=25, best calibrated=3.221e-03, exact=False, runtime=82s
(5600.6894854168131)*((phi_na + 0.020037220352414488*v*(-0.3009830237596048*phi_k - phi_na - 0.028295299061511524*phi_t)) - 0.46441502997715645*(phi_k - 0.13511758719540384*phi_t))*1.06829278450244)

seed 2: close; NRMSE=9.989e-03, R²=0.99985283, complexity=31, best calibrated=9.850e-03, exact=False, runtime=77s
(5600.6894854168131)*(phi_na + (phi_k - 2.3654128446919946)*(-0.41950955037457005) - 0.0637841319374151*(-phi_na + v*(phi_k*0.30878950287452706 + phi_na)*0.33994405484781165 - (-1.2440859606808121)*(phi_k - phi_t) - 1.2275998308037372) - 1*1.0739497597646048)

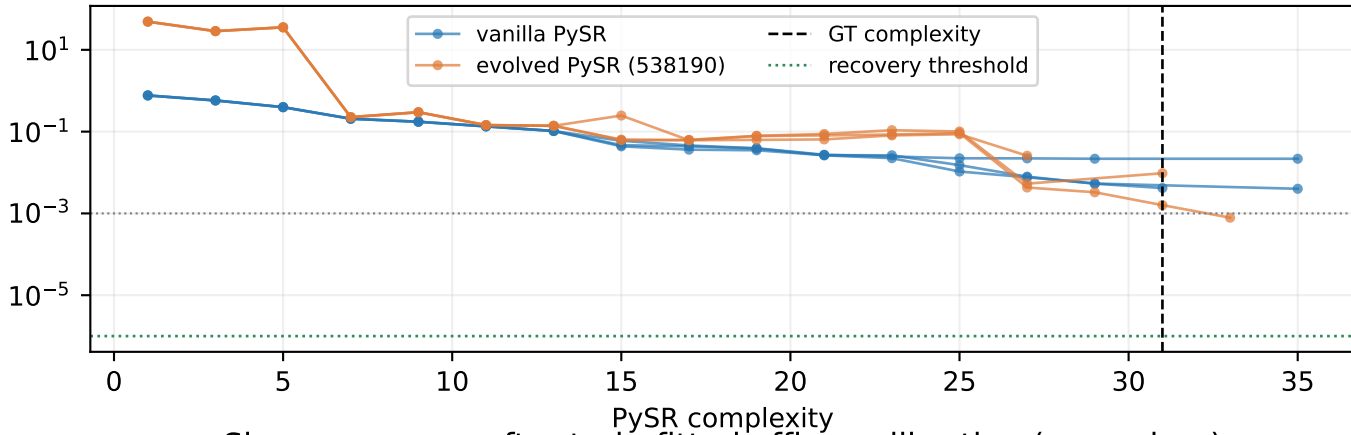
d_type

D-type K, de-inactivated by hyperpolarisation -> delayed/suppressed firing on the test

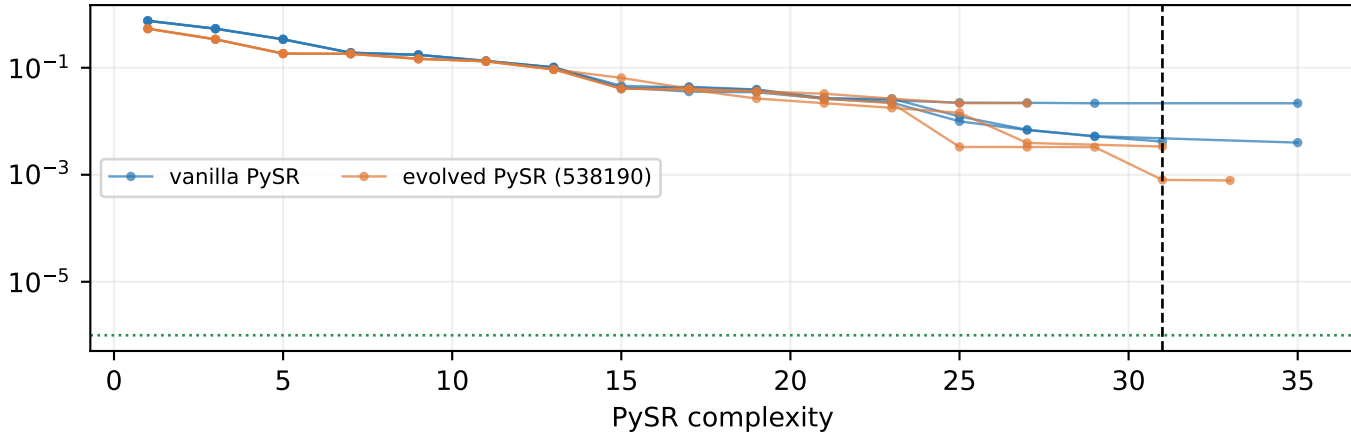
Ground truth (reference complexity 31):

$i_{ext} - (120) * \phi_{na} * (v - (50)) - (36) * \phi_k * (v - (-77)) - (0.29999999999999999) * (v - (-54.399999999999999)) - (9) * \phi_d * (v - (-77))$

Discovered Pareto frontiers (all seeds)



Shape recovery after train-fitted affine calibration (secondary)



vanilla PySR

seed 0: close; NRMSE=4.173e-03, R²=0.99997692, complexity=31, best calibrated=4.163e-03, exact=False, runtime=65s
 $(5428.7713685998297) * (\phi_{na} - (\phi_k * 0.5061729999748338 - 0.12929327232210053 * (-\phi_d + \phi_{na}) + (\phi_{na} + v) * (-(-0.006551760856253743) * \phi_k + (\phi_{na} + (\phi_d + 0.10157115424671317) * 0.07281152882032232) * 0.022047799616329758) + 0.009042250969408071))$

seed 1: close; NRMSE=4.005e-03, R²=0.99997874, complexity=35, best calibrated=3.990e-03, exact=False, runtime=67s
 $(5428.7713685998297) * (\phi_k * (-1.541760899106991) + \phi_k + \phi_{na} - 0.126937935787804 * (\phi_d - (\phi_d * (-0.012326411446044365) * v - 1 * (-0.14186055973121559))) + (\phi_{na} + (\phi_k - 1 * (-0.021987501184081725)) * 0.30370114658614544) * (v * (-0.021987501184081725) - 1 * (-0.11002291250234238)) - 0.027957351122344962)$

seed 2: close; NRMSE=2.169e-02, R²=0.99937658, complexity=35, best calibrated=2.168e-02, exact=False, runtime=65s
 $(5428.7713685998297) * (\phi_{na} - 0.5099034684346586 * (\phi_k + ((\phi_d + 0.5046104443604654) * 0.04122748032886335 - (v * (\phi_k * 0.2983044873295211 + \phi_{na} + 0.04079689319037041) + 0.5318380206715028)) * (-0.04335575057217011)) + (-\phi_d + \phi_{na} - 0.013185032721464427) * 0.10210018015774545 - 1 * 0.004296519428073108)$

evolved PySR (538190)

seed 0: close; NRMSE=2.569e-02, R²=0.99912516, complexity=27, best calibrated=2.170e-02, exact=False, runtime=78s
 $(5428.7713685998297) * ((\phi_{na} - 0.373822647795649 * (\phi_k + \phi_{na} * 0.0537802821034261 * v - (-0.24586006118015916) * (\phi_d + \phi_k + (\phi_k - 1 * (-0.12438591889056923)) * v * 0.06568761550694688))) * 1.0898240499652414)$

seed 1: close; NRMSE=5.336e-03, R²=0.99996225, complexity=27, best calibrated=3.366e-03, exact=False, runtime=76s
 $(5428.7713685998297) * ((-\phi_d * (v * (-0.0015310394682974825) - 0.11601291903267467) + \phi_{na} + (\phi_k - (-3.3205891464187354) * \phi_{na}) * (-\phi_{na} + v) * (-0.006188657025040012) - (\phi_k * 0.47385185379316364 - \phi_{na})) * 1.0813448183978207)$

seed 2: near-exact; NRMSE=7.862e-04, R²=0.99999918, complexity=33, best calibrated=7.829e-04, exact=False, runtime=75s
 $(5428.7713685998297) * ((\phi_d * (v * (-0.0015310394682974825) - 0.11601291903267467) + \phi_{na} + (\phi_k - (-0.043311519840647184) * \phi_{na} * v) * (-0.4627058276950213) + (i_{ext} * (-0.027564640668462207) + \phi_k * v) * (-0.006031607790788499)) * 1.1047595551024083 - 0.0021223773069490737)$

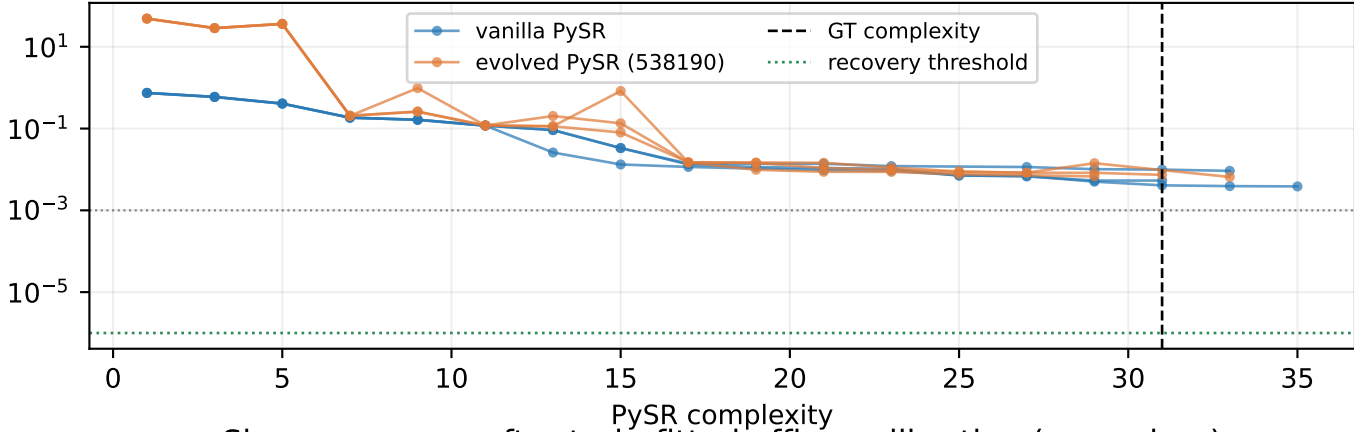
textbook M

TEXTBOOK M-current adaptation (Na+K+M): recallable -- names do all the work

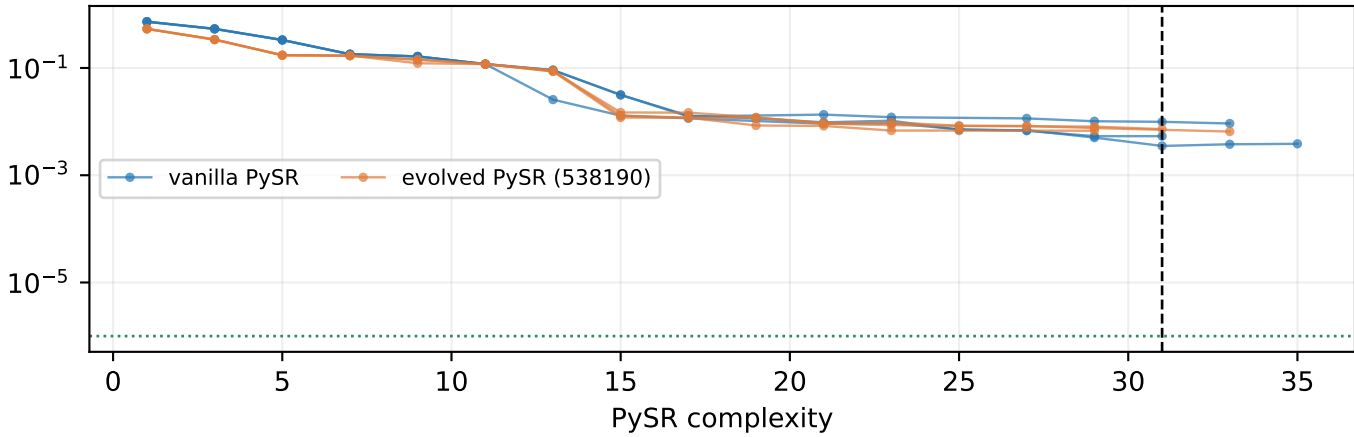
Ground truth (reference complexity 31):

$i_{\text{ext}} - (120) \cdot \phi_{\text{na}}(v - (50)) - (36) \cdot \phi_k(v - (-77)) - (0.29999999999999999) \cdot (v - (-54.399999999999999)) - (2.5) \cdot \phi_m(v - (-77))$

Discovered Pareto frontiers (all seeds)



Shape recovery after train-fitted affine calibration (secondary)



vanilla PySR

seed 0: close; NRMSE=5.353e-03, R²=0.99996007, complexity=31, best calibrated=5.353e-03, exact=False, runtime=65s
(5427.246966177755)*(-0.5048972300110478*(phi_k - phi_na) + (phi_na - 0.0342949149205068)*0.6220019972900185 + (phi_k*0.30154829720848225 - (phi_m*(-0.02090080139989714) - phi_na))*(phi_m + phi_m + v + 0.0005047240784157171)*(-0.02215578171222505))

seed 1: close; NRMSE=3.849e-03, R²=0.99997936, complexity=35, best calibrated=3.516e-03, exact=False, runtime=66s
(5427.246966177755)*(-(phi_k - 0.02215302104613533*(v*(-0.021577111729817897)*phi_m - phi_m - phi_na*(phi_m - 0.4887656406047185) - v*(phi_k*0.3011052882715898 + phi_na)) - (phi_k*(-0.4887656406047185) + phi_na*(-1.1048115576676747)) - 0.010650918643378122)

seed 2: close; NRMSE=9.232e-03, R²=0.99988124, complexity=33, best calibrated=9.219e-03, exact=False, runtime=60s
(5427.246966177755)*(-0.6710294019510215*phi_k + phi_na - 0.009458572614780699*(phi_na + phi_na)*(phi_m + phi_m + v - 2.3056053955364284) + (v - 22.667401394173403)*(phi_k + phi_k + phi_na)*(-0.0034386798917260303) - 0.017769415909336752)

evolved PySR (538190)

seed 0: close; NRMSE=6.758e-03, R²=0.99993636, complexity=29, best calibrated=6.757e-03, exact=False, runtime=79s
(5427.246966177755)*(phi_na*((v + 0.4930898403913758)*(-0.043260828824600875) - 1*(-3.1848534255165597)) - phi_na - (phi_k - (-0.053050500827910456)*phi_m) - ((phi_k + 0.04169576827493152)*v*0.013017798927790715 - 1*(-0.017745242189427232)))*0.5110958979710888)

seed 1: close; NRMSE=6.548e-03, R²=0.99994025, complexity=33, best calibrated=6.525e-03, exact=False, runtime=74s
(5427.246966177755)*(((phi_k + phi_m)*0.06454705659985085 - (-phi_k + phi_na + phi_na - 0.04044656611885704*(phi_k*0.3158685643243652 + phi_na)*(-(-0.03138892516928046)*phi_k*v*(-phi_k + phi_m) + v)))*(-0.5512298382866325))

seed 2: close; NRMSE=7.375e-03, R²=0.99992422, complexity=31, best calibrated=7.242e-03, exact=False, runtime=79s
(5427.246966177755)*((phi_m + phi_na - v + 3.0655391351089074)*(phi_k + phi_na*3.333927057773386 - 1*(-0.04421409884285506))*0.00663361150421018 + (-phi_k + phi_m*(-0.07762089114517269) - phi_na + phi_na*2.866639831643114)*0.5371076227009065)

Paired comparison: points below diagonal favor evolved PySR

